
Preface

This is a two-quarter long sequence covering point

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Chapter 1

Introduction

For the remainder of this chapter: let $U \subset \mathbb{R}^n$ be an open set. An element $x \in \mathbb{R}^n$ will always be of the form $(x_1, \dots, x_n)$. The function $u : U \rightarrow \mathbb{R}$ will always denote the unknown function being solved for in a differential equation. We denote $u := u(x) = u(x_1, \dots, x_n)$.

§ 1.1. Basic Definitions

Definition 1.1.1. The unit vector in $\mathbb{R}^n$ is denoted by e^i , where it is 1 in the i^{th} slot and 0 otherwise.

Definition 1.1.2. The partial derivative of u with respect to x_i is:

$$u_{x_i} := \frac{\partial u}{\partial x_i} := \lim_{h \rightarrow 0} \frac{u(x + he^i) - u(x)}{h}.$$

Definition 1.1.3. The gradient of u is the vector:

$$Du := \nabla u := (u_{x_1}, \dots, u_{x_n}).$$

Definition 1.1.4. The Hessian matrix of u is the 2 dimensional array:

$$D^2u = \begin{pmatrix} u_{x_1 x_1} & \dots & u_{x_1 x_n} \\ \vdots & \ddots & \vdots \\ u_{x_n x_1} & \dots & u_{x_n x_n} \end{pmatrix}$$

Definition 1.1.5. The Laplacian of u is:

$$\Delta u := \nabla^2 u := \text{Tr}(D^2u) = \sum_{i=1}^n u_{x_i x_i}.$$

Definition 1.1.6.

- (a) A vector of the form $\alpha := (\alpha_1, \dots, \alpha_n)$, where $\alpha_i \in \mathbb{Z}_{\geq 0}$, is called a multiindex of order $|\alpha|$, where:

$$|\alpha| = \alpha_1 + \dots + \alpha_n.$$

- (b) Given a multiindex α , we define:

$$D^\alpha u(x) := \frac{\partial^{|\alpha|} u(x)}{\partial^{\alpha_1} x_1 \dots \partial^{\alpha_n} x_n}.$$

Definition 1.1.7. A partial differential equation (PDE for short) is an equation involving an unknown function u along with one or more of its partial derivatives. A partial differential equation is linear if we can write it as:

$$\sum_{|\alpha| \leq K} a_\alpha(x) D^\alpha u(x) = f(x).$$

If $f(x) = 0$, then this PDE is called homogenous.

§ 1.2. Multivariable Calculus

Theorem 1.2.1 (Fubini). If $u \in L^1(U)$, then:

$$\int_U u dx = \underbrace{\int \dots \int}_{n\text{-times}} u(x_1, \dots, x_n) dx_1 \dots dx_n.$$

Definition 1.2.1. The boundary of U is denoted ∂U .

Definition 1.2.2. We say the boundary ∂U is $\underline{C^k}$ if in a neighborhood of any point on the boundary, the boundary is the graph of a C^k function (possibly after the coordinate system has been rotated).

Definition 1.2.3. If ∂U is C^1 , then along ∂U is defined the outward pointing unit normal vector field, which we denote by:

$$\vec{v} = (v^1, \dots, v^n).$$

Definition 1.2.4. Let $u \in C^1(\bar{U})$. The outward normal derivative of u is:

$$\frac{\partial u}{\partial v} := Du \cdot \vec{v}.$$

Definition 1.2.5. Let $\vec{u} = (u^1, \dots, u^n)$ be a vector field. The flux of u across ∂U is:

$$\int_{\partial U} \vec{u} \cdot \vec{v} dS.$$

Theorem 1.2.2 (Gauss-Green).

(a) Suppose $u \in C^1(\bar{U})$. For any $i = 1, \dots, n$, we have:

$$\int_U u_{x_i} dx = \int_{\partial U} u v^i dS$$

(b) (Divergence Theorem) Let $\vec{v} = (v^1, \dots, v^n)$, where $v^i \in C^1(\bar{U})$ for each $i = 1, \dots, n$. Then:

$$\int_U \operatorname{div} \vec{v} dx = \int_{\partial U} \vec{v} \cdot \vec{v} dS.$$

Theorem 1.2.3 (Integration by Parts). *Let $u, v \in C^1(\bar{U})$ and let ∂U be C^1 . Then:*

$$\int_U u_{x_i} v dx = - \int_U u v_{x_i} dx + \int_{\partial U} u v \nu^i dS.$$

Theorem 1.2.4 (Green's Formulas). *Let $u, v \in C^1(\bar{U})$ and let ∂U be C^1 . Then:*

$$\begin{aligned} (a) \quad & \int_U \Delta u dx = \int_{\partial U} \frac{\partial u}{\partial \nu} dS, \\ (b) \quad & \int_U Dv \cdot Du dx = - \int_U u \Delta v dx + \int_{\partial U} \frac{\partial u}{\partial \nu} u dS, \\ (c) \quad & \int_U u \Delta v - v \Delta u dx = \int_{\partial U} u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{\partial \nu} dS. \end{aligned}$$

§ 1.3. Polar Coordinated in $\mathbb{R}^n$

Remark. In $\mathbb{R}^n$, for any $x \neq 0$, we can write:

$$x = |x| \cdot \frac{x}{|x|} := r\omega,$$

where $r \in (0, \infty)$ and $\omega \in S^{n-1} = \partial B(0, 1)$.

Theorem 1.3.1. *There exists a measure $d\omega$ on S^{n-1} such that:*

$$\int_{\mathbb{R}^n} u dx = \int_0^\infty \left(\int_{S^{n-1}} u(r\omega) r^{n-1} d\omega \right) dr.$$

In other words, $dx = r^{n-1} d\omega dr$.

Corollary 1.3.1. *For any $x_0 \in \mathbb{R}^n$ and $a > 0$, we have:*

$$\int_{B(x_0, a)} u dx = \int_{B(0, a)} u(x_0 + y) dy = \int_0^a \left(\int_{S^{n-1}} u(x_0 + r\omega) r^{n-1} d\omega \right) dr.$$

Corollary 1.3.2. *For any $x_0 \in \mathbb{R}^n$ and $r > 0$, we have:*

$$\frac{d}{dr} \int_{B(x_0, r)} u dx = \int_{\partial B(x_0, r)} u dS.$$

§ 1.4. Smoothing

Definition 1.4.1.

- (a) Let $U, V \subset \mathbb{R}^n$. We say U is compactly contained in V , and write $U \subset\subset V$, if $V \subset \bar{V} \subset U$ and $\bar{V}$ is compact.

- (b) If $f|_V \in L^p(V)$ for all $V \subset\subset U$, then we say f is locally integrable on U . The set of all such functions is denoted $L^p_{\text{loc}}(U)$.

Definition 1.4.2. For $\epsilon > 0$, we define:

$$U_\epsilon := \{x \in U \mid \text{dist}(x, \partial U) > \epsilon\}.$$

Definition 1.4.3.

- (a) Let $x \in \mathbb{R}^n$. The standard mollifier is the function:

$$\eta(x) := \begin{cases} \frac{-C}{1-|x|^2}, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases},$$

where C is chosen so that $\int_{\mathbb{R}^n} \eta(x) dx = 1$.

- (b) For all $\epsilon > 0$, define:

$$\eta_\epsilon(x) := \frac{1}{\epsilon^n} \eta\left(\frac{x}{\epsilon}\right).$$

Proposition 1.4.1. We have $\eta \in C_c^\infty(\mathbb{R}^n)$. Moreover, for all $\epsilon > 0$, we have:

- (a) $\eta_\epsilon \in C_c^\infty(\mathbb{R}^n)$;
- (b) $\text{supp } \eta_\epsilon = B(0, \epsilon)$;
- (c) $\int_{\mathbb{R}^n} \eta_\epsilon(x) dx = 1$.

Definition 1.4.4.

- (a) For $\epsilon > 0$, we define:

$$U_\epsilon := \{x \in U \mid \text{dist}(x, \partial U) > \epsilon\}.$$

- (b) If $f \in L^p_{\text{loc}}(U)$, we define it mollifier as:

$$\begin{aligned} f^\epsilon(x) &:= (\eta_\epsilon * f)|_{U_\epsilon} \\ &= \int_U \eta_\epsilon(x-y) f(y) dy \\ &= \int_{B(0, \epsilon)} \eta_\epsilon(y) f(x-y) dy. \end{aligned}$$

Chapter 2

Solutions to Linear Partial Differential Equations

§ 2.1. The Transport Equation

§ 2.2. Laplace's Equation

Definition 2.2.1. Laplace's equation is:

$$\Delta u = 0.$$

A C^2 function satisfying Laplace's equation is called a harmonic function.

Proposition 2.2.1. *Laplace's equation is rotation invariant. Specifically, let $A = (a_{ij})$ denote an $n \times n$ orthogonal matrix. For $x \in \mathbb{R}^n$, let $y = Ax$; that is:*

$$Ax = y = (y_1, \dots, y_n), \quad y_i = \sum_{j=1}^n a_{ij} x_j$$

Assume $u \in C^2(\mathbb{R}^n)$ satisfies $\Delta u = 0$. Then define

$$v(x) = u(Ax) = u(y).$$

We have that $\Delta v = 0$.

Proof. Note that $\frac{\partial y_k}{\partial x_i} = a_{ki}$. Hence:

$$\begin{aligned} v_{x_i}(x) &= \frac{\partial}{\partial x_i} u(y) \\ &= \sum_{k=1}^n u_{y_k}(y) \frac{\partial y_k}{\partial x_i} \\ &= \sum_{k=1}^n u_{y_k}(y) a_{ki}. \end{aligned}$$

Taking a second partial derivative with respect to x_i gives:

$$\begin{aligned} v_{x_i x_i}(x) &= \frac{\partial}{\partial x_i} \sum_{k=1}^n u_{x_k}(y) a_{ki} \\ &= \sum_{k=1}^n \sum_{j=1}^n u_{x_k x_j}(y) \frac{\partial y_j}{\partial x_i} a_{ki} \\ &= \sum_{j=1}^n \sum_{k=1}^n a_{ji} a_{ki} u_{x_j x_k}. \end{aligned}$$

Summing from $i = 1, \dots, n$ finally yields:

$$\begin{aligned} \Delta v(x) &= \sum_{i=1}^n v_{x_i x_i}(x) \\ &= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n a_{ji} a_{ki} u_{x_j x_k}(y) \\ &= \sum_{j=1}^n \sum_{k=1}^n \left(\sum_{i=1}^n a_{ji} a_{ki} \right) u_{x_j x_k}(y) \\ &= \sum_{j=1}^n \sum_{k=1}^n \delta_{jk} u_{x_j x_k}(y) \\ &= \sum_{k=1}^n u_{x_k x_k} \\ &= \Delta u(y) \\ &= 0 \end{aligned}$$

■

Example 2.2.1. Since solutions to Laplace's equation are rotation invariant, we are interested in finding *radial solutions* to the equation; i.e., functions of $r = |x|$. Let u be a solution of Laplace's equation and suppose further we have $u(x) = v(r)$. Then $u_{x_i} = v'(r)r_{x_i}$ and $u_{x_i x_i} = v''(r)r_{x_i}^2 + v'(r)r_{x_i x_i}$. Using the fact that:

$$\begin{aligned} \frac{\partial}{\partial x_i} r^2 &= \frac{\partial}{\partial x_i} \sum_{i=1}^n x_i^2 \\ &= 2r r_{x_i}, \end{aligned}$$

we get that $r_{x_i} = \frac{x_i}{r}$. A similar calculation yields $r_{x_i x_i} = \frac{1}{r} - \frac{x_i^2}{r^3}$. Now since u is a solution to Laplace's equation, we have that $\Delta u = 0$. We can write this as:

$$\sum_{i=1}^n u_{x_i x_i} = 0.$$

Substituting for our earlier calculation gives:

$$\sum_{i=1}^n v''(r) \frac{x_i^2}{r^2} + v'(r) \left(\frac{1}{r} - \frac{x_i^2}{r^3} \right) = 0.$$

Hence:

$$v''(r) + v'(r) \frac{(n-1)}{r} = 0.$$

For $n = 2$, the solution to the above ODE is $v(r) = c_1 \log r + c_2$. For $n > 2$, the solution to the above ODE is $v(r) = c_1 \frac{1}{r^{n-1}} + c_2$.

Definition 2.2.2. The fundamental solution to Laplace's equation is:

$$\Phi(x) = \begin{cases} c_2 \log \frac{1}{|x|}, & n = 2 \\ c_n |x|^{2-n}, & n > 2, \end{cases}$$

where $c_2 = \frac{1}{2\pi}$ and $c_n = \frac{1}{n(n-2)\alpha(n)}$, where $\alpha(n)$ denotes the volume of $B(0, 1)$ in $\mathbb{R}^n$.

Remark. The fundamental solution to Laplace's equation admits a singularity at $x = 0$.

Remark. If u and v are both harmonic, then $au + bv$ is harmonic for any $a, b \in \mathbb{R}$ since the Laplacian is a linear differential operator. Moreover, if u is harmonic then $u(x - x_0)$ is also harmonic when treated as a variable with respect to x . These reasons combined lead to the following conclusion: if $y^1, \dots, y^k \in \mathbb{R}^n$ and $f_1, \dots, f_k \in \mathbb{R}$, the function:

$$\sum_{i=1}^k f_i \Phi(x - y^i)$$

is also harmonic! A natural question to ask now is whether:

$$\int_{\mathbb{R}^n} f(y) \Phi(x - y) dy$$

is harmonic.

Theorem 2.2.1. Let $f \in C_c^2(\mathbb{R})$. Define $u(x) = \int_{\mathbb{R}^n} f(y) \Phi(x - y) dy$. Then $u \in C^2(\mathbb{R}^n)$ and $-\Delta u = f$.

Proof. Taking the difference quotient of $u(x)$ gives the following equality:

$$\frac{u(x + he^i) - u(x)}{h} = \int_{\mathbb{R}^n} \frac{f(x + he^i - y) - f(x - y)}{h} \Phi(y) dy.$$

Applying the Dominated Convergence Theorem gives:

$$u_{x_i} = \int_{\mathbb{R}^n} f_{x_i}(x - y) \Phi(y) dy.$$

Taking the difference quotient of both sides and applying the Dominated Convergence Theorem again yields:

$$u_{x_i x_i} = \int_{\mathbb{R}^n} f_{x_i x_i}(x - y) \Phi(y) dy.$$

Thus:

$$\begin{aligned}\Delta u &= \int_{\mathbb{R}^n} \Delta_x f(x-y) \Phi(y) dy \\ &= \int_{\mathbb{R}^n} (-1)^2 \Delta_y f(x-y) \Phi(y) dy.\end{aligned}$$

Since f is a function with compact support, let $U := B^o(0, R)$ be large enough such that $x \in U$ and $\text{supp } f(x-y) \subset U$. Our integral becomes:

$$\begin{aligned}\Delta u &= \int_{\mathbb{R}^n} \Delta_y f(x-y) \Phi(y) dy \\ &= \int_U \Delta_y f(x-y) \Phi(y) dy.\end{aligned}$$

Recall that our fundamental solution Φ admits a discontinuity at the origin. For an arbitrary $\epsilon > 0$, we have:

$$\Delta u = \int_{U \setminus B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) dy + \int_{B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) dy.$$

Since $\Delta_y f(x-y) \Phi(y)$ does not admit a discontinuity on $U \setminus B(0, \epsilon)$, we can apply Green's identities. Taking advantage of the fact that $\Delta_y \Phi(y) = 0$, we have:

$$\begin{aligned}\int_{U \setminus B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) dy &= \int_{U \setminus B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) + f(x-y) \Delta_y \Phi(y) dy \\ &= \int_{\partial(U \setminus B(0, \epsilon))} \frac{\partial f}{\partial \nu}(x-y) \Phi(y) - f(x-y) \frac{\partial \Phi}{\partial \nu}(y) dS(y).\end{aligned}$$

Integrating on $\partial(U \setminus B(0, \epsilon))$ is equivalent to:

$$\int_{\partial U} \frac{\partial f}{\partial \nu}(x-y) \Phi(y) - f(x-y) \frac{\partial \Phi}{\partial \nu}(y) dS(y) - \int_{\partial B(0, \epsilon)} \frac{\partial f}{\partial \nu}(x-y) \Phi(y) - f(x-y) \frac{\partial \Phi}{\partial \nu}(y) dS(y).$$

The outward normal derivative of f is zero, and since f is of compact support it is zero along the boundary of U . The left-most integral is equal to zero, resulting in:

$$\Delta u = \underbrace{\int_{B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) dy}_{:=I} - \underbrace{\int_{B(0, \epsilon)} \frac{\partial f}{\partial \nu}(x-y) \Phi(y) dy}_{:=K} - \underbrace{\int_{\partial B(0, \epsilon)} f(x-y) \frac{\partial \Phi}{\partial \nu}(y) dy}_{:=L}$$

■

Definition 2.2.3. The equation:

$$-\Delta u = f$$

arising from Theorem 2.2.1 is called Poisson's equation.

Definition 2.2.4. Let $U \subset \mathbb{R}^n$ and $f \in L^1(\mathbb{R}, \mu)$. The average of f over U is:

$$\oint f d\mu := \frac{1}{\mu(U)} \int_U f d\mu.$$

Theorem 2.2.2 (Mean-Value Property). *Let $U \subset \mathbb{R}^n$ and $x \in U$. Suppose $u \in C^2(U)$ and $\Delta u = 0$ in U . For each $0 < r < \text{dist}(x, \partial U)$, we have:*

$$u(x) = \int_{B(x,r)} u(y) dy = \int_{\partial B(x,r)} u dS.$$

Proof. Let $x \in U$. For each $0 < r < \text{dist}(x, \partial U)$, define:

$$\begin{aligned} \rho(r) &:= \int_{\partial B(x,r)} u dS \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x,r)} u dS \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{S^{n-1}} u(x + r\omega) r^{n-1} d\omega \\ &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} u(x + r\omega) d\omega \end{aligned}$$

Taking the derivative of ρ with respect to r gives:

$$\begin{aligned} \rho'(r) &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} \frac{d}{dr} u(x + r\omega) d\omega \\ &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} Du(x + r\omega) \cdot \omega d\omega. \end{aligned}$$

Note that ω is precisely the outward normal on S^{n-1} . Making the substitution, $y = x + r\omega$ we have that $dy = r^{n-1} d\omega$ and $\omega = \frac{y-x}{r}$. Hence:

$$\begin{aligned} \rho'(r) &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} Du(x + r\omega) \cdot \omega d\omega \\ &= \frac{1}{n\alpha(n)r^{1-n}} \int_{\partial B(0,1)} Du(y) \cdot \frac{y-x}{r} dy. \end{aligned}$$

The final integrand is by definition the outward normal derivative of $u(y)$. Green's identity yields:

$$\begin{aligned} \rho'(r) &= \frac{1}{n\alpha(n)r^{1-n}} \int_{B(0,1)} \Delta u(y) dy \\ &= 0 \end{aligned}$$

Thus $\rho(r)$ is constant. With this fact, notice that.

$$\begin{aligned} \rho(r) &= \lim_{r \rightarrow 0^+} \rho(r) \\ &= u(x). \end{aligned}$$

This establishes $u(x) = \oint_{\partial B(x,r)} u \, dS$. Finally:

$$\begin{aligned}
 \oint_{B(x,r)} u(y) \, dy &= \frac{1}{\alpha(n)r^n} \int_{B(x,r)} u(y) \, dy \\
 &= \frac{1}{\alpha(n)r^n} \int_0^r \int_{S^{n-1}} u(x + \omega t) t^{n-1} \, d\omega \, dt \\
 &= \frac{1}{\alpha(n)r^n} \int_0^r t^{n-1} \int_{S^{n-1}} u(x + \omega t) \, d\omega \, dt \\
 &= \frac{1}{\alpha(n)r^n} \int_0^r n\alpha(n)t^{n-1} u(x) \, dt \\
 &= u(x). \quad \blacksquare
 \end{aligned}$$

Theorem 2.2.3. *If $u \in C^2(U)$ satisfies $u(x) = \oint_{\partial B(x,r)} u \, dS$ for every $B(x,r) \subset U$, then $\Delta u = 0$.*

Proof. ■

Theorem 2.2.4. *Let $U \subset \mathbb{R}^n$. Let $u \in C(U)$. If u satisfies:*

$$u(x) = \oint_{\partial B(x,r)} u \, dS = \oint_{B(x,r)} u(y) \, dy,$$

then $u \in C^\infty(U)$.

Proof. ■

Corollary 2.2.1. *Let $U \subset \mathbb{R}^n$. If u satisfies $\Delta u = 0$ in U , then $u \in C^\infty(U)$.*

Proof. ■

Theorem 2.2.5 (Maximum Principle). *Let $U \subset \mathbb{R}^n$ and $u \in C^2(U) \cap C(\overline{U})$. Assume $\Delta u = 0$ in U . Then:*

- (a) (Weak Maximum Principle) $\max_{\overline{U}} u = \max_{\partial U} u$;
- (b) (Strong Maximum Principle) *If U is connected, and there exists $x_0 \in U$ such that $u(x_0) = \max_{\overline{U}} u$, then u is constant on U .*

Proof. (a) As $\partial U \subseteq \bar{U}$, we immediately have $\max_{\partial U} u \leq \max_{\bar{U}} u$. For the other direction of inequality, let $\epsilon > 0$. Consider the function $u + \epsilon|x|^2$. Then:

$$\begin{aligned} \Delta(u + \epsilon|x|^2) &= \Delta u + \epsilon \Delta|x|^2 \\ &= \epsilon \sum_{j=1}^n \frac{\partial}{\partial x_j} \left(\frac{\partial}{\partial x_j} |x|^2 \right) \\ &= \epsilon \sum_{j=1}^n \frac{\partial}{\partial x_j} 2x_j \\ &= \epsilon \sum_{j=1}^n 2 \\ &= 2n\epsilon \\ &> 0. \end{aligned}$$

If $u + \epsilon|x|^2$ were to attain a local maximum at an interior point, then $\frac{\partial^2}{\partial x_j^2} (u + \epsilon|x|^2) \leq 0$ for all $j = 1, \dots, n$. This immediately implies that $\Delta(u + \epsilon|x|^2) \leq 0$, contradicting the above result. Since the local maximum of $u + \epsilon|x|^2$ must lie on its boundary, we have:

$$\begin{aligned} \max_{\bar{U}} u &\leq \max_{\bar{U}} (u + \epsilon|x|^2) \\ &= \max_{\partial U} (u + \epsilon|x|^2) \end{aligned}$$

Taking $\epsilon \rightarrow 0$ gives $\max_{\bar{U}} u \leq \max_{\partial U} u$, establishing the claim.

(b) Assume $u(x_0) = \max_{\bar{U}} u := M$. Let $B(x_0, r) \subset U$. By the Mean Value Property:

$$\begin{aligned} M &= u(x_0) \\ &= \int_{B(x_0, r)} u(x) dx \\ &= \frac{1}{\alpha(n)r^n} \int_{B(x_0, r)} u(x) dx \\ &\leq \frac{M}{\alpha(n)r^n} \int_{B(x_0, r)} dx \\ &= M. \end{aligned}$$

If the maximum of $B(x_0, r)$ is $u(x_0)$, and if $\int_{B(x_0, r)} u(x) dx = M$, it must be the case that $u(x) = M$ on all of $B(x_0, r)$. Since u is locally constant on a connected space, it is globally constant; i.e., $u(x) = M$ for all $x \in U$. ■

Remark. Note that the strong maximum principle implies the weak maximum principle.

Remark. A similar theorem can be stated in terms of local minima of functions, and the proof follows by consider $-u(x)$ and applying the maximum principle.

Corollary 2.2.2. Let $y \in C(\partial U)$ and $f \in C(U)$. Solutions to the boundary value problem:

$$\begin{cases} \Delta u = f, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases}$$

are unique.

Proof. Let u and v be solutions to the above boundary value problem. Since the Laplacian is linear, this induces the following boundary value problem:

$$\begin{cases} \Delta(u - v) = 0, & \text{in } U \\ u - v = 0, & \text{on } \partial U. \end{cases}$$

Since $u - v$ is harmonic in U , Theorem 2.2.5 and Theorem ?? assert that $\max_U(u - v) = \min_U(u - v)$. It must be the case that $u - v = 0$ everywhere on U , hence $u = v$. ■

Theorem 2.2.6 (Interior Derivate Estimate). Assume $\Delta u = 0$ in U . Then:

$$|Du| \leq \frac{C}{r^{n+1}} \|u\|_{L^1(B(x_0, r))}$$

for each $B(x_0, r) \subset U$.

Proof. First, note that if u is a harmonic function then u_{x_i} is also harmonic (Clairaut's Theorem). As such, the function u_{x_i} admits the Mean-Value Property:

$$\begin{aligned} |u_{x_i}(x_0)| &= \left| \oint_{B(x_0, \frac{r}{2})} u_{x_i} dx \right| \\ &= \frac{2^n}{\alpha(n)r^n} \left| \int_{B(x_0, \frac{r}{2})} u_{x_i} dx \right| \\ &= \frac{2^n}{\alpha(n)r^n} \left| \int_{\partial B(x_0, \frac{r}{2})} u v^i dS \right| \\ &\leq \frac{2^n}{\alpha(n)r^n} \sup \left\{ |u(x)| : x \in \partial B\left(x_0, \frac{r}{2}\right) \right\} \int_{\partial B(x_0, \frac{r}{2})} dS \\ &= \frac{2^n}{\alpha(n)r^n} \|u\|_{L^\infty(\partial B(x_0, \frac{r}{2}))} n\alpha(n) \frac{r^{n-1}}{2^{n-1}} \\ &= \frac{2n}{r} \|u\|_{L^\infty(\partial B(x_0, \frac{r}{2}))} \end{aligned}$$

Note that for any $x \in \partial B(x_0, \frac{r}{2})$, we have $B(x, \frac{r}{2}) \subset B(x_0, r)$. Using this fact we can estimate

$|u(x)|$ as follows:

$$\begin{aligned}
 |u(x)| &= \left| \int_{B(x, \frac{r}{2})} u(y) dy \right| \\
 &= \frac{2^n}{\alpha(n)r^n} \int_{B(x, \frac{r}{2})} |u(y)| dy \\
 &= \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x, \frac{r}{2}))} \\
 &\leq \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0, r))}.
 \end{aligned}$$

Taking the supremum over all $x \in \partial B(x_0, \frac{r}{2})$ gives:

$$\|u\|_{L^\infty(\partial B(x_0, \frac{r}{2}))} \leq \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0, r))}.$$

Combining this with our first string of inequalities results in:

$$\begin{aligned}
 |u_{x_i}(x_0)| &\leq \frac{2n}{r} \cdot \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0, r))} \\
 &= \frac{n2^{n+1}}{\alpha(n)r^{n+1}} \|u\|_{L^1(B(x_0, r))}.
 \end{aligned}$$

Defining $C := \frac{n2^{n+1}}{\alpha(n)}$ and taking the gradient of $u_{x_i}(x_0)$ gives the desired result. $\blacksquare$

Corollary 2.2.3 (Liouville). *Suppose $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is harmonic. If u is bounded, then u is constant.*

Proof. Let $x_0 \in \mathbb{R}^n$ and $r > 0$. Assume $|u(x)| < M$. For each $i = 1, \dots, n$ we have:

$$\begin{aligned}
 |u_{x_i}(x_0)| &\leq \frac{C}{r^{n+1}} \|u\|_{L^1(B(x_0, r))} \\
 &\leq \frac{C}{r^{n+1}} \int_{B(x_0, r)} M dx \\
 &\leq \frac{CM}{r^{n+1}} \alpha(n)r^n \\
 &= \frac{CM\alpha(n)}{r}.
 \end{aligned}$$

As we are assuming u is harmonic on all of $\mathbb{R}^n$, taking $r \rightarrow \infty$ does not fail any of the necessary assumptions made in Theorem 2.2.6. Since $u_{x_i}(x_0) = 0$, we conclude that u is constant. $\blacksquare$

Theorem 2.2.7. *Assume $\Delta u = 0$ in U . Then:*

$$|D^\alpha u(x_0)| \leq \frac{C_k}{r^{n+k}} \|u\|_{L^1(B(x_0, r))}$$

for each $B(x_0, r) \subset U$ and each multiindex α of order $|\alpha| = k$.

Proof.

